

Stat 5200: Analysis of Designed Experiments

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Overview

Objectives and overview

- This material is based Chapter 3 on the textbook “A First Course in Design and Analysis of Experiments” (Oehlert2000).
- Chapter outcomes include:
 - TODO

Completely Randomized Designs

Completely Randomized Design

- Completely randomized design (CRD) is the simplest design for comparing treatments and is sometimes the preferred design.
- Setup:
 - There are g treatment groups.
 - N experimental units (EU)s available.
 - You want sample sizes of $n_1, n_2, \dots, n_g$, with $\sum_{i=1}^g n_i = N$.

Completely Randomized Design II

- **Idea:** All possible sub-group composition are equally as likely.

Performing a Completely Randomized Design

- Randomly choose n_1 EU's for treatment 1.
 - Randomly choose n_2 EU's for treatment 2.
 - ...
 - Randomly choose n_{g-1} EU's for treatment $g - 1$.
 - The remaining n_g EU's are assigned to treatment g .
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- On a computer, this can be done by permuting $1, 2, \dots, N$, and assigning the first n_1 numbers to treatment 1, the next n_2 to treatment 2, etc.

Advantages and Disadvantages

- Advantages
 - Simple to do; all you need is one random permutation.
 - Simple to explain and understand.
 - Simple to analyze.
 - No restrictions on the sample sizes (other designs may restrict n_i).
 - Easy with missing values.
- Disadvantages
 - Not always possible or cost-effective.
 - The accuracy of the experiment depends on variability among the EUs, not just the variability among treatment groups. Thus, it may require more EUs (\$\$\$\$).

References and Acknowledgements

Oehlert GW (2000). *A First Course in Design and Analysis of Experiments*. W. H. Freeman. ISBN 0-7167-3510-5.

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